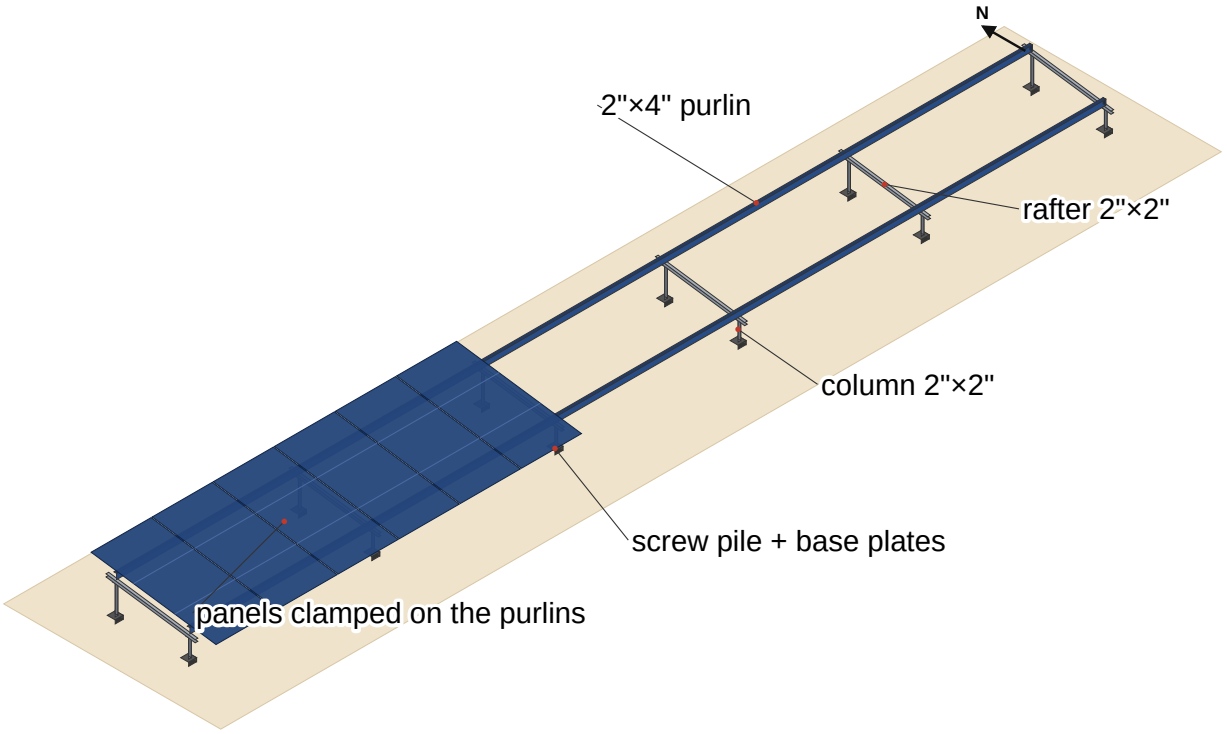
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INSTALLATION MANUAL

STR-02 Ground-Mount Structure — 45 Panels on 36 Screw Piles

Reference design · low ground mount for 45 × 650 W panels · Chiang Rai area



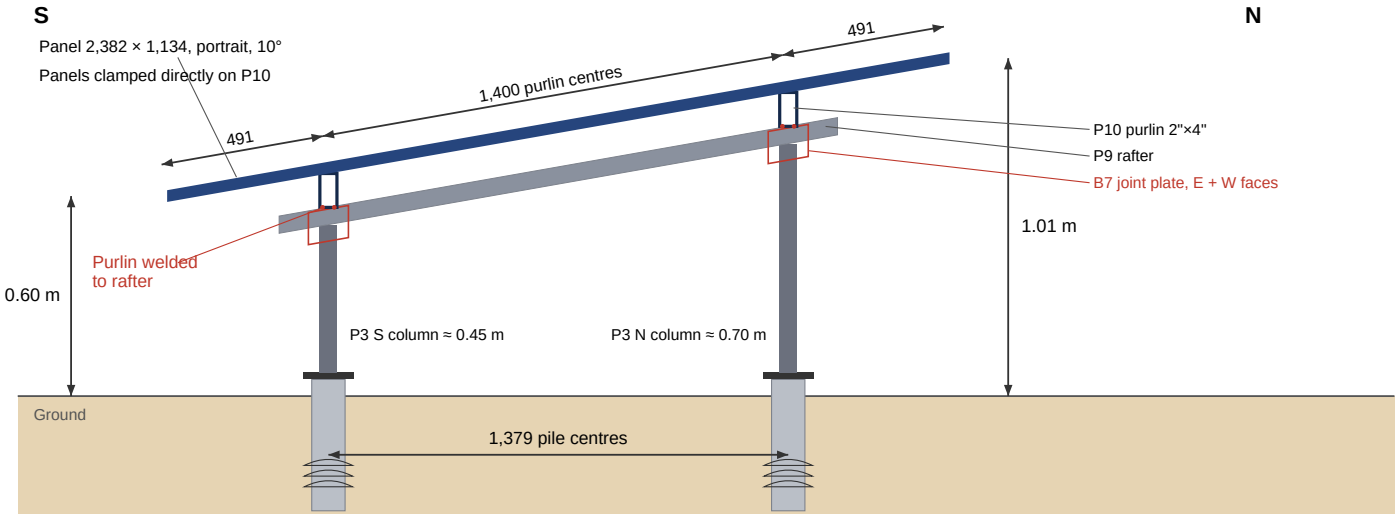
One row of 15 panels (6 shown) on 6 frames. Three rows make the full array.

Drawings	STR-02 drawings: Sheet 1 (footings), Sheet 2 (pile-to-column base), Sheet 3 (frame)
Array	3 rows × 15 panels, portrait, 10° facing south; panel edges 0.60 m (south) and 1.01 m (north)
Foundations	36 screw piles Ø 100 mm × 1.5 m, 6 per purlin line at 3.46 m; head plate 50 mm above ground
Frame	18 frames: 2"×2"×2.3 columns and rafters; 2"×4"×2.3 purlins (2 per row); panels clamped directly on the purlins; joint plates at every rafter; knee braces at F1, F3, F6
Document	MAN-STR02 · Rev04 · 11 November 2026 · issued by Solar ESS Working Group

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1. System overview

Each support frame stands on two screw piles: a south column and a north column, joined at the top by a sloping rafter at 10°. A 2"×4" purlin runs east–west along the top of each line of columns, welded to every rafter. The panels are clamped straight onto the two purlins of their row. There are no concrete footings or rails. Each rafter-to-column joint has two joint plates, and knee braces on frames F1, F3 and F6 keep each row stiff east–west.



Section through one frame, looking west (Sheet 3, E5.1).

2. Parts

Ref	Part	Qty
A1/A2	Screw pile Ø 100 × 1,500 mm, 2.0 mm wall, HDG, with 150 × 150 head plate (4 slots Ø 15 × 30)	36
B1	Column base plate 150 × 150 × 10 mm, 4 slots 14 × 30	36
B2	Gusset 40 × 60 × 5 mm (column base)	144
B3–B5	M12 × 60 gr 8.8 HDG bolt + nut; plate washers 30 × 30 × 3 (2 per bolt); M12 spring washer	144 sets
B6	Levelling shims 150 × 150 × 1 / 2 / 3 mm	≈ 20
B7	Joint plate 120 × 100 × 5 mm, rafter-to-column (east and west faces, 2 per joint)	72
C1–C2	2"×2"×2.3 tube for columns (≈ 0.45 / 0.70 m) and rafters (1.70 m)	10 × 6 m
C3	2"×4"×2.3 tube for purlins (6 lines: 6.0 + 5.3 + 6.0 m)	18 × 6 m
C4	Backing strips for purlin butt welds, flat bar 40 × 3 × 100 mm (2 per joint, 12 joints)	24
C5	Knee braces, 2"×2"×2.3 tube ≈ 0.38 m, ends cut at 45° (F1, F3, F6; 4 per frame)	36 (3 × 6 m)
D1–D4	Mid clamps; end clamps; M8 clamp bolts; M8 rivet nuts	84; 12; 96; 96
E1–E5	Zinc-rich paint; tube end caps 2"×2" and 2"×4"; E6013 electrodes; earthing lugs	3 L; 36 + 12; 7 kg; 3

2a. Design data and wind resistance

Design wind
25 m/s
DPT 1311-50, Zone 1 reference speed (1-hour mean, 50-year return); gusts about 35 m/s (127 km/h)

Within allowable stress up to
≈ 28 m/s
reference speed; gusts about 39 m/s (140 km/h). Limited by the purlin under downward wind

Steel first yields at about
≈ 36 m/s
reference speed; gusts about 50 m/s (≈ 180 km/h)

Capacities are calculated from the member checks below by scaling wind pressure with the square of wind speed. They are engineering estimates, not a type test.

Design load	Value	Basis
Reference velocity pressure q	0.39 kPa (40 kgf/m²)	½ ρ V², ρ = 1.25 kg/m³, V = 25 m/s
Net wind uplift on panels	1.20 kPa (122 kgf/m²)	Open ground-mounted array, 10° tilt, gust factor included
Downward design pressure	1.45 kPa (148 kgf/m²)	Downward wind 1.27 kPa + dead load 0.18 kPa
Dead load (panels + steel)	≈ 2.2 t in total; 0.18 kPa	45 × 28.5 kg panels (1,283 kg) + ≈ 905 kg steel
Uplift per row (15 panels, 40.5 m²)	≈ 5.04 t	8 × 504 kgf interior + 4 × 252 kgf end supports
Uplift per pile, interior / row end	504 / 252 kgf	Tributary 3.46 m × 1.19 m
Downward per pile, interior	≈ 610 kgf	1.45 kPa × 4.1 m²
Lateral per support	87 kgf	Applied at 1.0 m above ground (conservative)

Resistance check	Demand / capacity	Safety factor
Pile withdrawal (rated, depends on soil)	504 / 1,900 kgf	3.77 (row end 7.5); whole array 36 × 1,900 kg = 68 t vs ≈ 15.1 t uplift
Pile compression (rated)	610 / 3,800 kgf	≈ 6.2
Pile lateral	87 kgf	Site test at 130 kgf (1.5 ×), < 10 mm
Purlin 2"×4"×2.3, downward	114 / 141 MPa; sag ≈ 9 mm (L/385)	1.24 (governing member)
Purlin, uplift	81 / 141 MPa	1.74
Column 2"×2"×2.3	157 / 235 MPa	1.5
Base plate 10 mm	219 / 235 MPa (before gussets)	1.07
Base bolts M12 gr 8.8	6.6 / ≈ 48 kN per bolt	> 7
Purlin-to-rafter weld	4.9 / ≈ 27 kN	≈ 5.5
Panel clamps (uplift per mid clamp)	≈ 1.62 kN	Use clamps and rivet nuts rated ≥ 2 kN pull-out
Panel glass, rear (uplift)	1.20 kPa	Check against the panel datasheet rear load rating

Other numbers	Value	Note
Array	45 × 650 W = 29.25 kWp	121.6 m² of panels on 17.3 m × 8.6 m of ground
Tilt / azimuth / lowest edge	10° / south / 0.60 m	North edge 1.01 m
Earthquake	≈ 61 kg dead load per pile	Horizontal earthquake force is well below the 87 kgf wind lateral load
Thermal movement of a 17.3 m purlin	≈ 8 mm for a 40 °C swing	Taken up by column flexibility and the 20 mm panel gaps
Corrosion	HDG piles and bolts; zinc-rich paint on welds	Inspect yearly; touch up paint every 2–3 years

3. Tools and safety

Tools

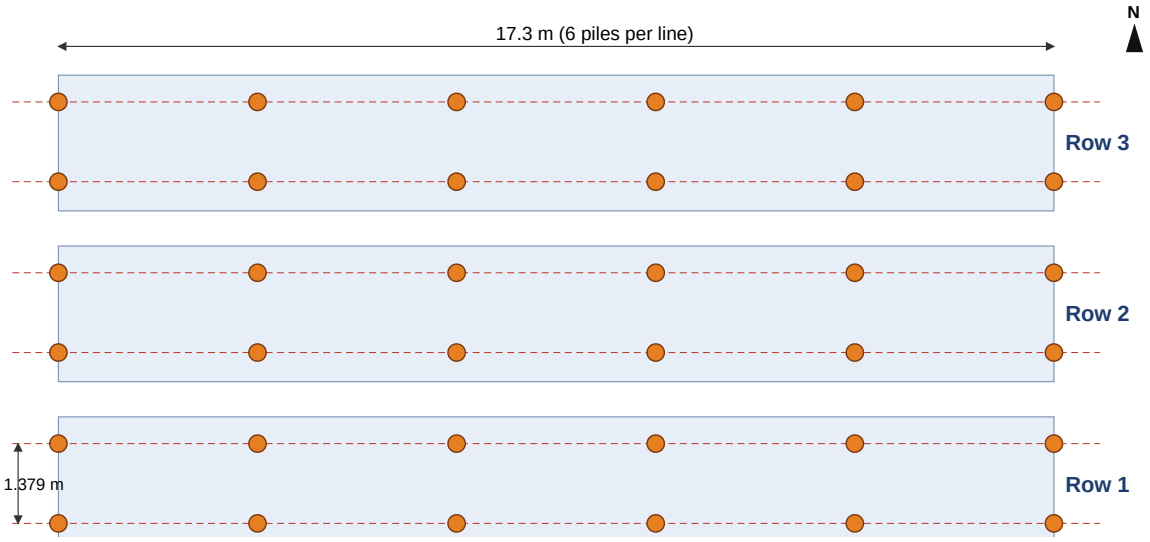
- Screw-pile driver with torque read-out, extension and adaptor
- Pull-out jack with frame and dial gauge; spring balance or load cell for the lateral test
- Laser level or water level, string lines, 30 m tape, spirit level
- Arc welder, E6013 2.6 mm electrodes, grinder with cutting and flap discs
- Drill with Ø 9 / 11 / 13 mm bits, M8 rivet-nut tool
- Torque wrench (25–80 N·m), spanners 13 / 17 / 19 mm

Safety

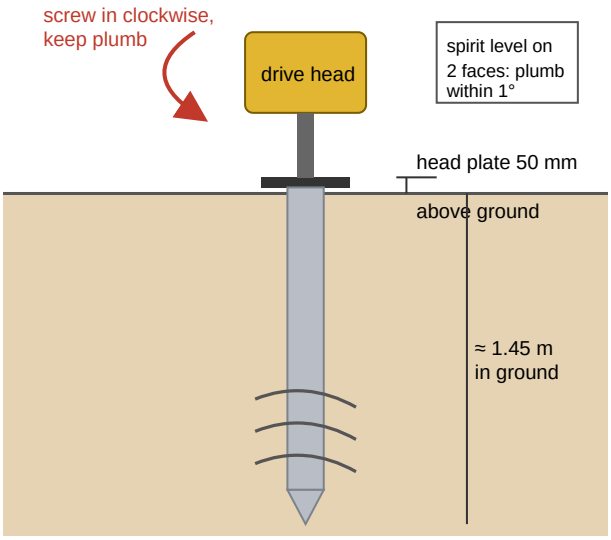
- Check for buried cables and pipes before setting out the piles
- Hard hat, safety shoes, gloves; welding mask and fire extinguisher at every welding point
- Keep hands clear of the pile driver while it turns; two people on the driver
- Paint welds only when cool; work in ventilated air
- Panels: lift by two people, never stand on panels, no work in strong wind or rain
- Keep the panel strings open-circuit until the system is commissioned

4. Step 1 — Set out the piles

1. Mark the array corners from the site plan (overall 17.3 m east–west × 8.6 m north–south, south at the bottom). Check the diagonals are equal.
2. Stretch one string line per pile line: two per row, 1.379 m apart north–south; row centres 3.11 m apart.
3. Mark 6 pile centres on each line at 3.46 m. Tolerance ± 20 mm along the line, ± 10 mm across it.

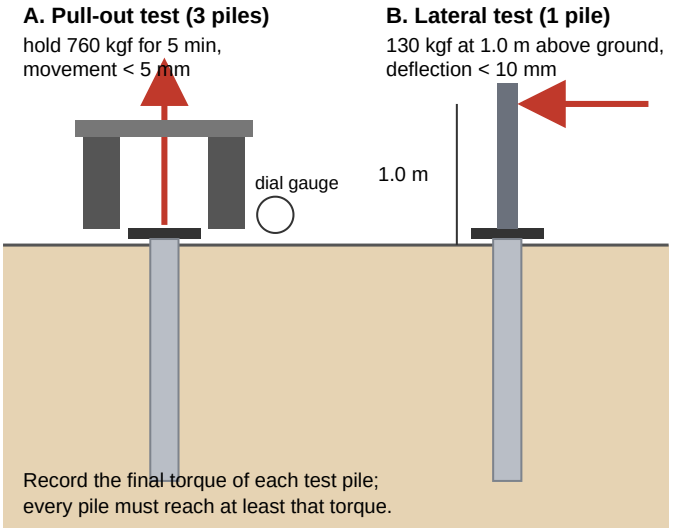


5. Step 2 — Install the screw piles



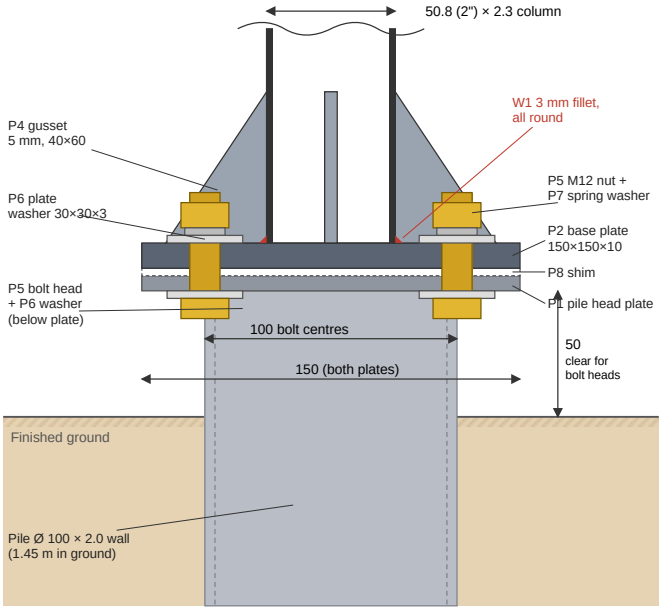
1. Start each pile on its mark with the driver, check it is plumb on two faces, then screw it in clockwise at a steady speed.
2. Stop when the head plate is 50 mm above finished ground (about 1.45 m in the ground). Do not back the pile out to adjust height; small differences are taken up when the columns are cut.
3. Turn the head plate so its slots lie on a diagonal of the row.
4. Write down the final torque of every pile.
5. If a pile stops early (rock or large stones), move it up to 100 mm along its line and try again. If it still stops, tell the site engineer: another pile or a concrete footing, designed by Solar ESS, is used at that support.

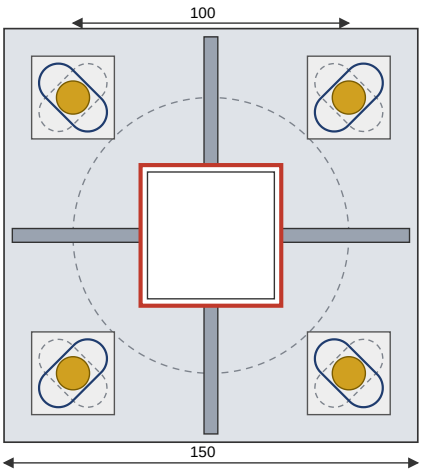
6. Step 3 — Site tests (before any panels)



1. **Pull-out:** on at least three piles spread over the site, pull up with 760 kgf and hold for 5 minutes. Movement must be under 5 mm.
2. **Lateral:** on one pile, with a short post bolted on, push sideways with 130 kgf at 1.0 m above ground. Deflection must be under 10 mm.
3. Note the final torque of the test piles. Every other pile must reach at least that torque.
4. A failed test is solved by moving or adding a pile, or by a concrete footing designed by Solar ESS.

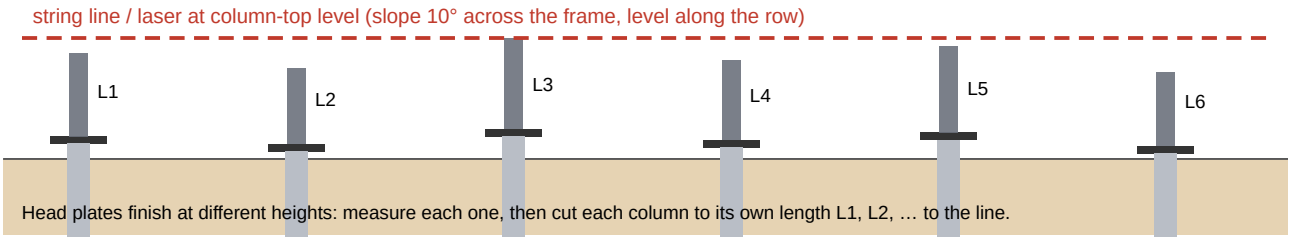
7. Step 4 — Base plates and columns





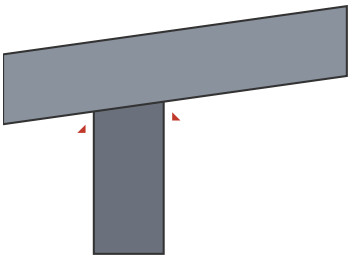
Plan on the base plate: base-plate slots at 90° to the pile slots give ± 7 mm adjustment both ways.

1. Measure the level of every head plate against the laser or string line.
2. Cut each column to its own length so its top meets the frame line: about 0.45 m (south) and 0.70 m (north). Cut the top at 10° to fit under the rafter.
3. Weld each column to the centre of a base plate (3 mm fillet all round) and add the 4 gussets.
4. Set the base plate on the head plate with shims if needed (6 mm total at most). Fit 4 × M12 × 60 bolts, heads below, with plate washers on both sides and a spring washer under the nut.
5. Line up the column on the string line, check it is plumb, then tighten the nuts to about 70 N·m.



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8. Step 5 — Rafters



Column top cut at 10°; rafter welded on site, 3 mm fillet all round; zinc-rich paint, 2 coats

1. Cut 18 rafters of 1.70 m from 2"×2"×2.3 tube.
2. Lay each rafter across its south and north column tops, projecting 150 mm past each column, at 10°.
3. Tack-weld, check the slope and the frame line, then weld all round with a 3 mm fillet.
4. Weld a joint plate (B7) on the east and west faces of every joint, half on the rafter and half on the column, 3 mm fillet all round. Fit end caps on the rafter ends.

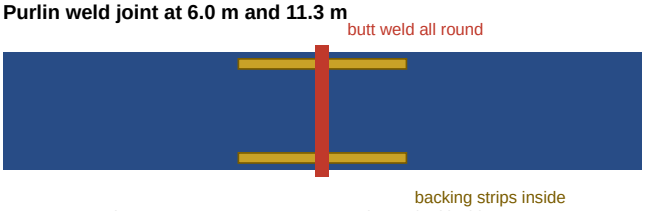
9. Step 6 — Purlins



Purlin 2"×4"×2.3 continuous over 6 frames; panels clamped on top · vertical scale × 2.2

Row elevation looking south: 6 frames at 3.46 m, purlin weld joints at 6.0 m and 11.3 m (vertical scale × 2.2).

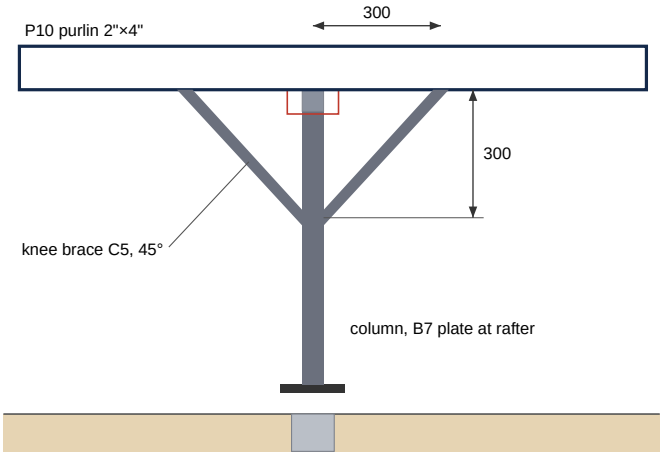
Purlin weld joint at 6.0 m and 11.3 m



Square-cut ends, 2 mm gap, two 40 × 3 × 100 strips tacked inside on the long sides; full-penetration weld all round, grind flush, paint.

1. Cut each purlin line from three 6 m lengths: 6.0 + 5.3 + 6.0 m.
2. Lay the 2"×4" purlin with its long side upright on the rafters, directly above the columns, from end frame to end frame.
3. Join the lengths at 6.0 m and 11.3 m from the west end: square-cut the ends, set a 2 mm gap with two backing strips tacked inside, then make a full-penetration butt weld all round and grind it flush. Keep the joint straight with a straight-edge across it.
4. Check the purlin is straight (± 5 mm over the row), then weld it to each rafter with a 3 mm fillet on both sides of the crossing.

Knee braces on frames F1, F3 and F6



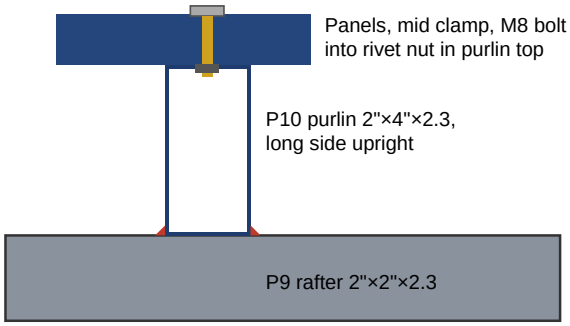
Elevation looking south at one braced column (both pile lines, both sides).

5. On frames F1, F3 and F6, weld two knee braces (C5) to every column: one east and one west, at 45°.
6. Each brace runs from the column face 300 mm below the purlin to the purlin underside 300 mm from the column centre. Cut both ends to fit, tack, check the purlin is still straight, then weld 3 mm all round.
7. 4 braces per frame, 12 per row, 36 in total. Paint all brace welds with two coats of zinc-rich paint.

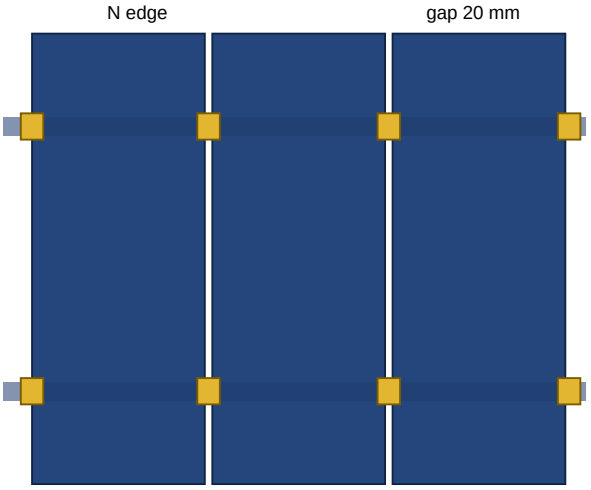
10. Step 7 — Corrosion protection and earthing

1. Chip and wire-brush every weld. Apply two coats of zinc-rich paint to all welds, cut ends and scratches.
2. Fit plastic end caps to all open tube ends (2"×2" rafters, 2"×4" purlins).
3. Bolt one earthing lug to a base plate in each row and connect it to the earth system.

11. Step 8 — Fix the panels



Purlin welded directly to the rafter: 3 mm fillet on both sides of the crossing (2 × 51 mm)



Yellow: clamps on the purlin lines (end clamps at row ends)

1. Mark the clamp positions on the top of each purlin: one at every panel joint and one at each row end (panel width 1,134 mm plus a 20 mm gap).
2. Drill Ø 11 mm (or as the rivet-nut maker says) and set an M8 rivet nut at each mark: 96 in total.
3. Lay the panels portrait, long sides north–south, with the purlins 491 mm from each panel end. Check this against the panel installation manual's clamping zone.
4. Fit the end clamps at the row ends and the mid clamps between panels. Tighten the M8 bolts to the clamp maker's torque (typically 14–18 N·m).
5. Check the panel edges finish at about 0.60 m (south) and 1.01 m (north) and that the row is straight.

12. Final inspection checklist

Item	Check	OK	Remarks
Pile positions	± 20 mm along the line, ± 10 mm across	☐	
Pile head plates	50 mm above ground; plumb within 1°; torque recorded	☐	
Site tests	3 pull-out (760 kgf, < 5 mm), 1 lateral (130 kgf at 1.0 m, < 10 mm)	☐	
Base bolts	4 per support, plate washers both sides, ≈ 70 N·m	☐	
Columns	Plumb within 1°; welds 3 mm all round; 4 gussets	☐	
Rafters	10° slope; 150 mm past each column; welds complete	☐	
Joint plates	2 per rafter-to-column joint (east and west), welded all round	☐	
Purlins	Straight ± 5 mm; welded at every rafter; butt welds at 6.0 / 11.3 m complete, ground and painted	☐	
Knee braces	F1, F3, F6: 4 per frame at 45°, 300 mm legs, welded all round	☐	
Corrosion	All welds and cut ends painted twice; end caps fitted	☐	
Earthing	One lug per row connected	☐	
Panels	Clamps at every joint and end; bolts at torque; edges 0.60 / 1.01 m	☐	

13. Notes

- This manual follows the STR-02 drawings, Sheets 1–3. If the manual and the drawings differ, the drawings govern.
- All steel is SS400. All bolts are grade 8.8, hot-dip galvanized.
- Any change to pile positions, footing type or member sizes on site must be recorded and confirmed by Solar ESS.

Installed by _____

Checked by (Solar ESS) _____

Date ____ / ____ / _____